COSC3106: Theory of Computing

CH#3:

Lec#9: DFA to RE

School of Computer Science and Technology

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This lecture

- In this lecture, you will …
  - Learn how to describe languages and presenting them in regular expressions.

- **Converting DFA to a regular expression:**
  - This means that to represent the language that to be accepted by a DFA in a RE.

- To convert from a DFA to RE, we need a tool that we can use to solve equation involving languages.

  - Lemma: Let Σ be an alphabet, and let $B, C,$ and L be languages in Σ^* and $\epsilon \notin B$, then:

    - If

$$L = BL \cup C. \qquad (1)$$

    - Then,

$$L = B^*C \qquad (2)$$

    - To prove this lemma, we need to solve for L to express it in terms of B and C

- Lemma: Let Σ be an alphabet, and B, C, and L be languages in Σ^* and $\epsilon \notin B$, then:
  - If $L = BL \cup C$ (1)
  - Then, $L = B^*C$ (2)

  - We need to solve (1) for L to express it in terms of B^* and C

Proof:

- If $u \in L \Rightarrow u \in BL \cup C$ so we have many options:

$$\Rightarrow u \in BL \quad or \quad u \in \cup C$$

  - Let us take: $u \in BL$ (3)

  - From (1), we have: $L = BL \cup C$, to be substituted in (3)

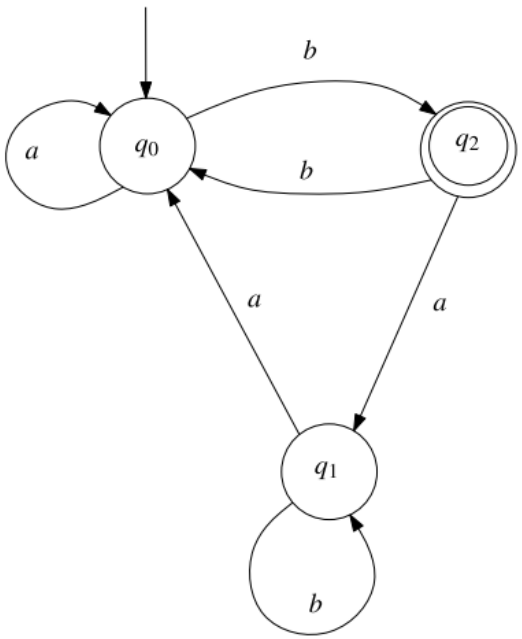
$$\Rightarrow u = BBL \; or \; BC$$

$$L = BL \cup C \Rightarrow u = BBBL \; or \; BBC$$

$$L = BL \cup C \Rightarrow u = BBBBL \; or \; BBBC$$

$\Rightarrow L = B^*C$ which proves that (1) = (2)

- Ex: Consider the DFA, $M = (Q, \Sigma, \delta, q_0, F)$, where $Q = \{q_0, q_1, q_2\}$, $\Sigma = \{a, b\}$, q_0 is the start state, $F = \{q_2\}$, and δ is given in the following state diagram below.



- We need to show how to obtain the regular expression that describes the language accepted by M.

Solution:

- For $i = 0, 1,$ and 2.
- We assume that we have 3 machines each has a start state q_i and has an unknown language L_i

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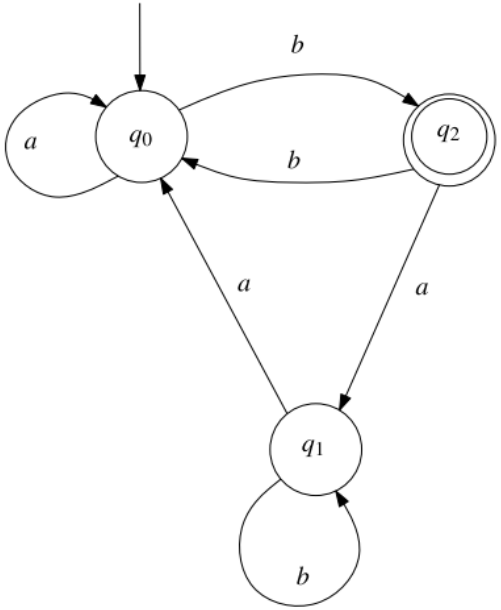
- Ex: (cont.): For $i = 0, 1,$ and 2. We assume that we have 3 machines each has a start state q_i and assume unknown strings L_i's
  - L_0: string in L_0 such that,

a	$\in L_0$

or

b	$\in L_2$

$$\Rightarrow L_0 = aL_0 \cup bL_2 \qquad (1)$$



  - L_1: string in L_1 such that,

a	$\in L_0$

or

b	$\in L_1$

$$\Rightarrow L_1 = aL_0 \cup bL_1 \qquad (2)$$

  - L_2: string in L_2 such that,

a	$\in L_1$

or

b	$\in L_0$

or ε

$$\Rightarrow L_2 = \varepsilon \cup aL_1 \cup bL_0 \qquad (3)$$

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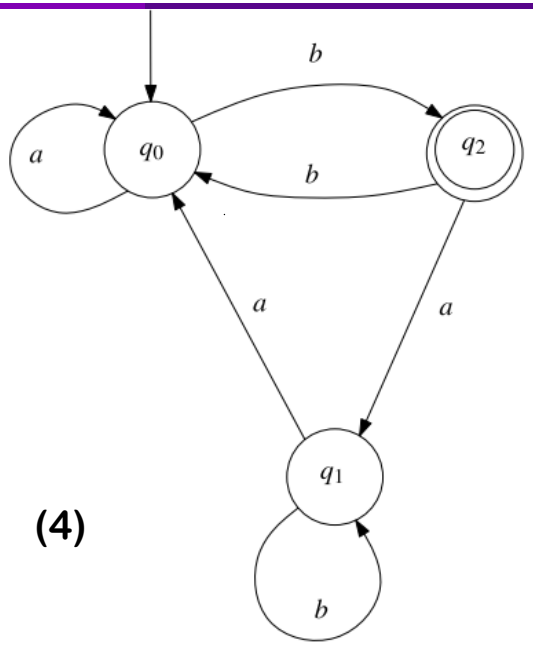
- Ex: (cont.): now we have,

$$L_0 = aL_0 \cup bL_2 \qquad (1)$$
$$L_1 = aL_0 \cup bL_1 \qquad (2)$$
$$L_2 = \varepsilon \cup aL_1 \cup bL_0 \qquad (3)$$



- Substituting (3) in (1):

$$\Rightarrow L_0 = aL_0 \cup b[\varepsilon \cup aL_1 \cup bL_0]$$
$$= aL_0 \cup b \cup baL_1 \cup bbL_0$$
$$= (a \cup bb)L_0 \cup b \cup baL_1 \qquad (4)$$

- From (2), we have

$$\underbrace{L_1}_{L} = \underbrace{bL_1}_{BL} \cup \underbrace{aL_0}_{C} \qquad (2)$$

$$\Rightarrow L_1 = b^*aL_0 \qquad (5)$$

$$L_0 = \left((a \cup bb) \cup ba\, b^*a\right)^* b$$

- Now, substitute (5) in (4)

$$\Rightarrow L_0 = (a \cup bb)L_0 \cup b \cup ba\, b^*aL_0$$
$$\underbrace{L_0}_{L} = \underbrace{\left((a \cup bb) \cup ba\, b^*a\right)}_{B}\underbrace{L_0}_{L} \cup \underbrace{b}_{C}$$

$L_0 = \left((a \cup bb) \cup ba\, b^*a\right)^* b$ is the RE of the DFA

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